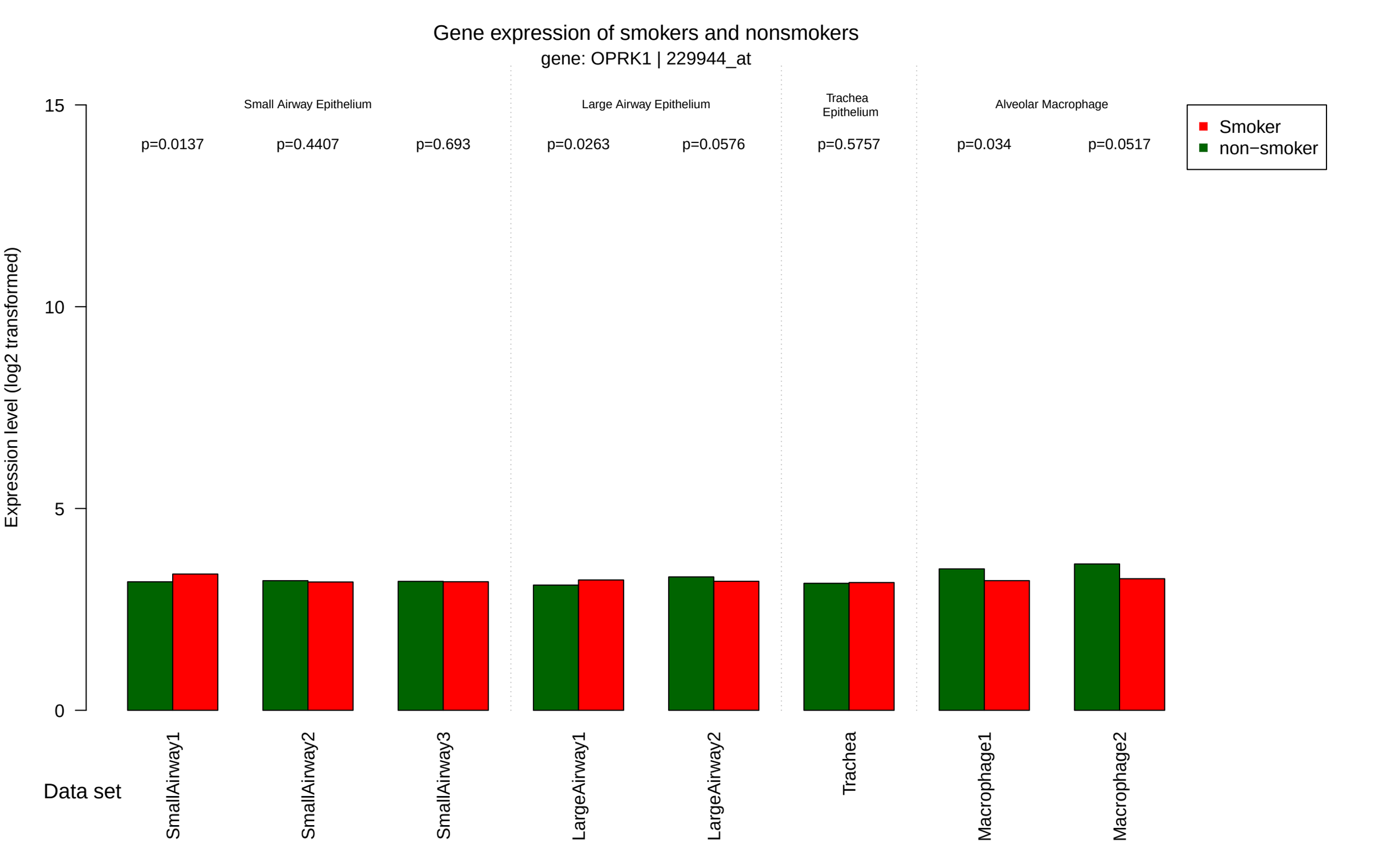
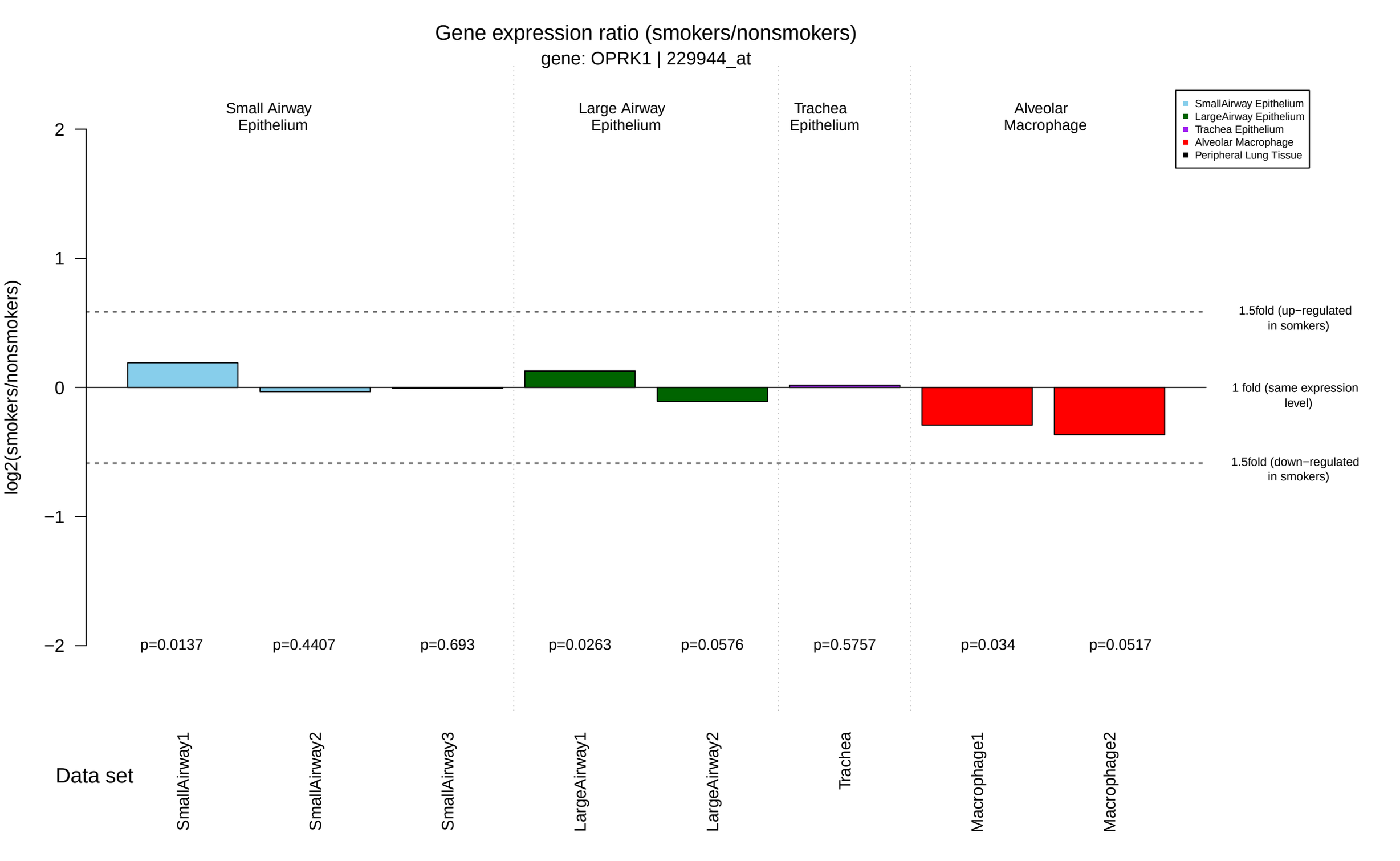
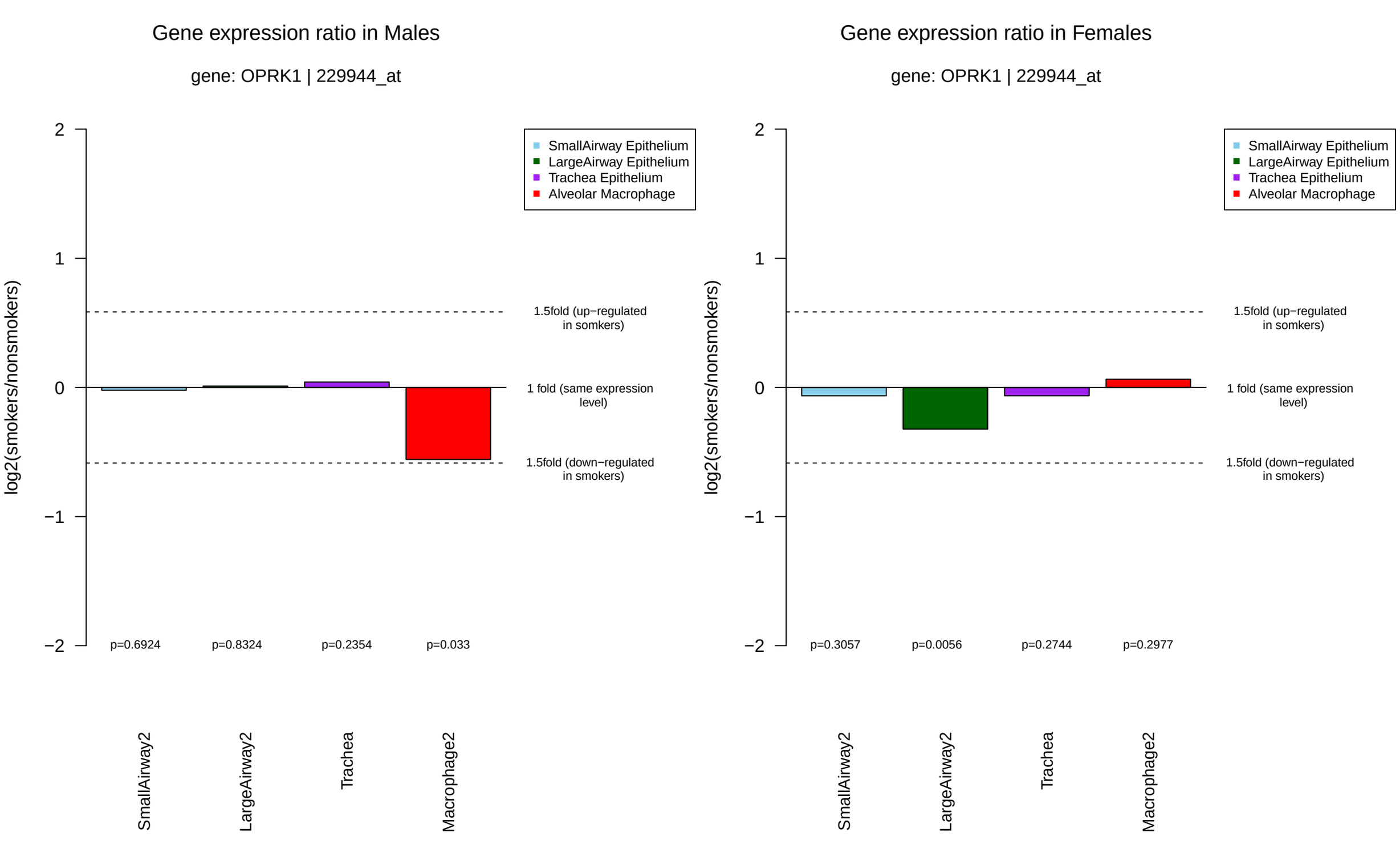
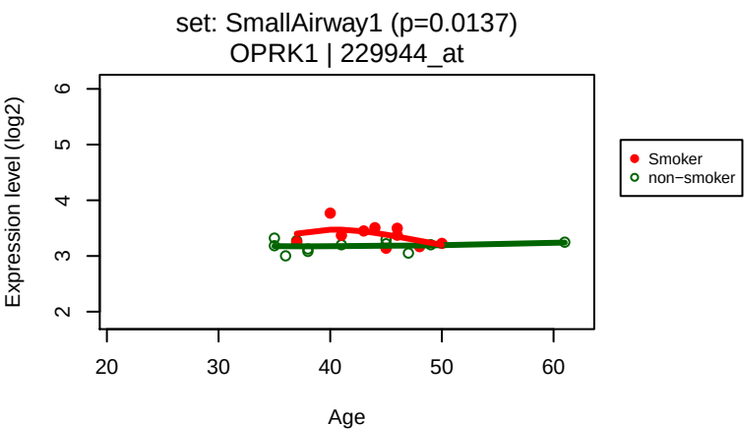
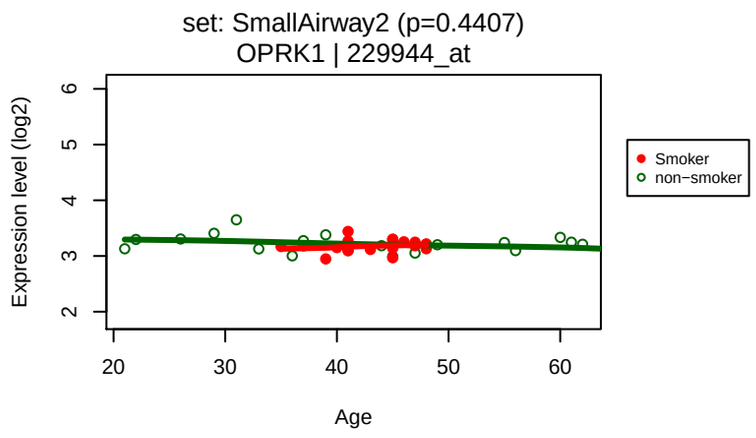
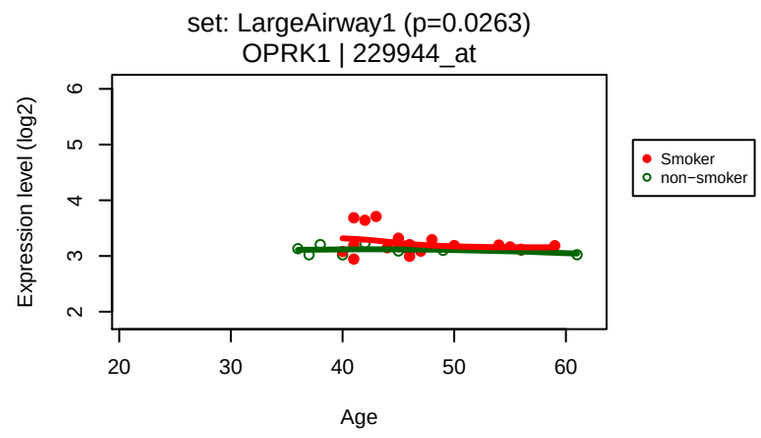

Gene expression of smokers and nonsmokers
gene: OPRK1 | 229944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0137
p=0.4407
p=0.693
p=0.0263
p=0.0576
p=0.5757
p=0.034
p=0.0517
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

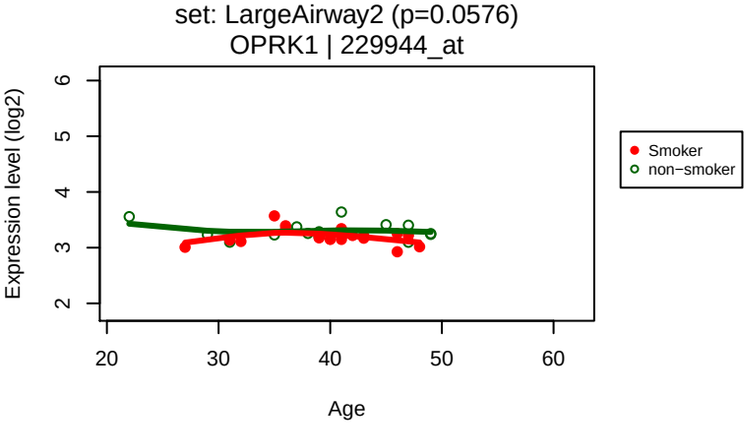

Gene expression ratio (smokers/nonsmokers)
gene: OPRK1 | 229944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0137
p=0.4407
p=0.693
p=0.0263
p=0.0576
p=0.5757
p=0.034
p=0.0517
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

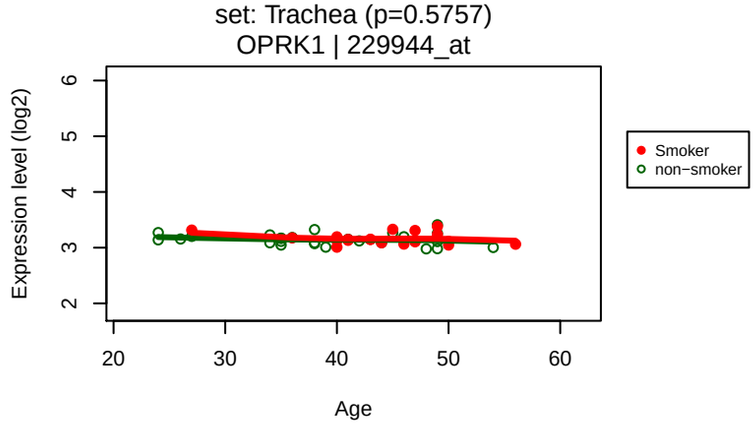

Gene expression ratio in Males
gene: OPRK1 | 229944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6924
p=0.8324
p=0.2354
p=0.033
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPRK1 | 229944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3057
p=0.0056
p=0.2744
p=0.2977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

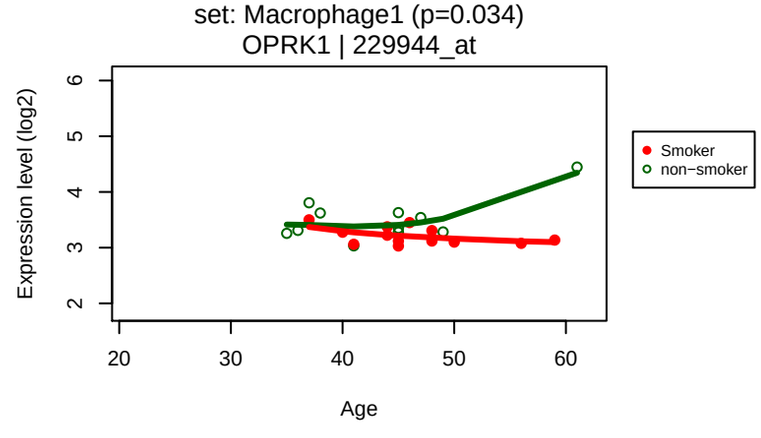

set: SmallAirway1 (p=0.0137)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker

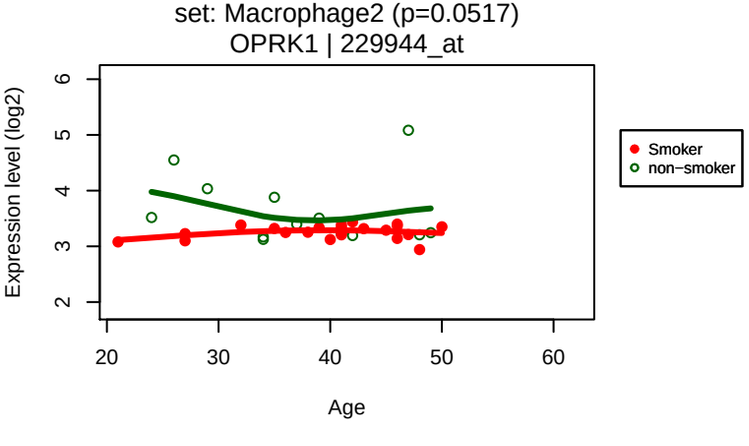

set: SmallAirway2 (p=0.4407)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker

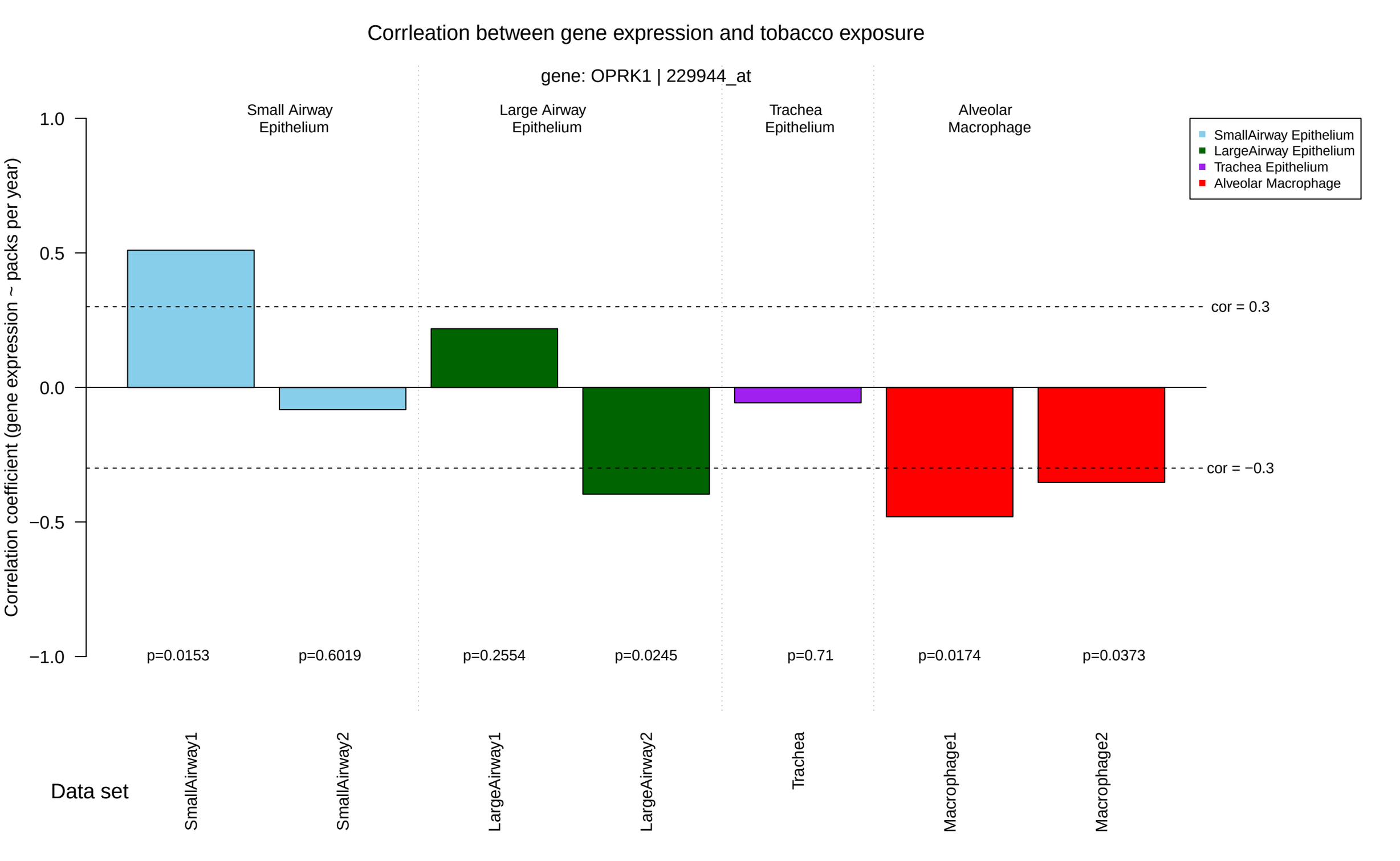

set: LargeAirway1 (p=0.0263)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0576)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5757)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.034)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0517)
OPRK1 | 229944_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPRK1 | 229944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0153
p=0.6019
p=0.2554
p=0.0245
p=0.71
p=0.0174
p=0.0373
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2